SCOB032 COMPUTER NETWORKS

Master Examination Study Guide — Version 2

With Original Textbook Figures & Teaching Diagrams

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Official assessment scope 2016–2026 + Tanenbaum 5e + Forouzan 5e + lecture slides

This version adds the missing figure extraction and labelled teaching diagrams required by the instructional design brief.

Figure Policy (as required by the Master Academic Study Guide brief)

Original textbook figures are extracted from Tanenbaum 5e at high resolution and kept identifiable as textbook sources. Simplified teaching diagrams are clearly labelled “SIMPLIFIED TEACHING DIAGRAM — BASED ON THE ORIGINAL TEXTBOOK FIGURE / LECTURE MATERIAL”. They preserve every academically essential component, label, relationship and direction; only non-essential visual complexity is removed. Technical terminology is never replaced by inaccurate everyday substitutes.

1. Original Textbook Figure — OSI Reference Model

Source: Tanenbaum & Wetherall, Computer Networks, 5th edition, Figure 1-20.

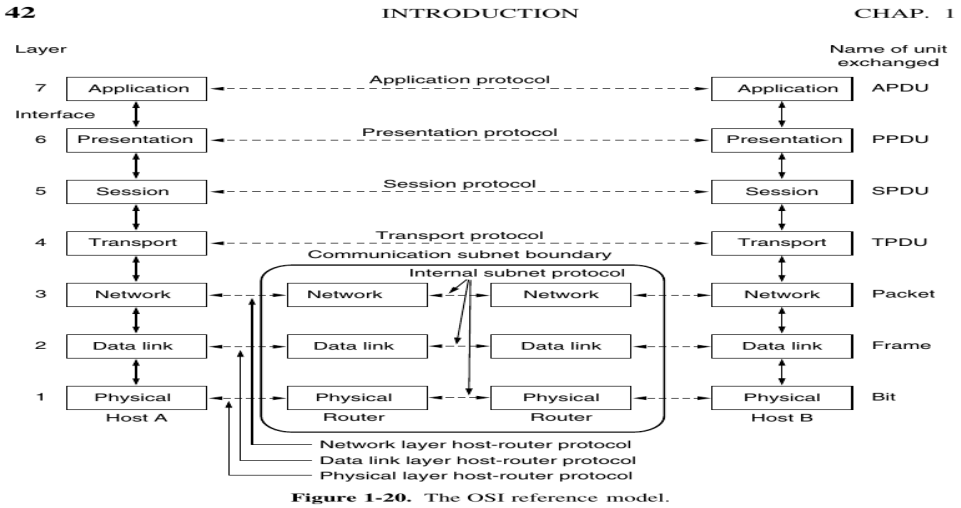


Figure 1-20. The OSI reference model.

4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

Below we will discuss each layer of the model in turn, starting at the bottom layer. Note that the OSI model itself is not a network architecture because it does not specify the exact services and protocols to be used in each layer. It just tells what each layer should do. However, ISO has also produced standards for all the layers, although these are not part of the reference model itself. Each one has been published as a separate international standard. The *model* (in part) is widely used although the associated protocols have been long forgotten.

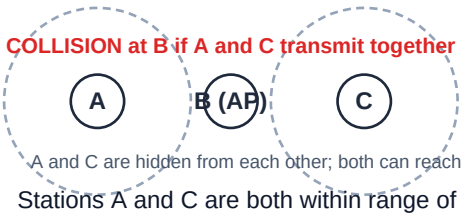
Original textbook page containing Figure 1-20 — The OSI reference model (Tanenbaum 5e).

What it represents: The seven layers of the OSI reference model and the units exchanged at each layer (bits, frames, packets, TPDUs, etc.). Host A and Host B communicate peer-to-peer at each layer; intermediate routers operate only up to the Network layer.

Examination relevance: You must be able to identify which layer is responsible for a given symptom (weak signal → Layer 1; wrong application → Layer 4; cannot leave subnet → Layer 3; MAC delivery inside subnet → Layer 2).

2. Hidden Terminal Problem & RTS/CTS

SIMPLIFIED TEACHING DIAGRAM — BASED ON TEXTBOOK / LECTURE MATERIAL (Tanenbaum Ch. 4, lecture slides MOBILE NETWORKS & Datalink layer)



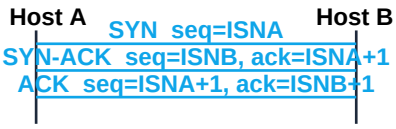
Stations A and C are both within range of B (the access point) but outside each other's radio range. If A and C transmit at the same time, their frames collide at B even though neither A nor C detects the collision. This is why CSMA/CD cannot be used on wireless media.

RTS/CTS solution: A sends a short RTS; B replies with CTS; every station that hears the CTS (including C) remains silent for the duration of A's subsequent data frame. The hidden terminal is silenced without requiring A and C to hear each other.

Student trap (past papers): "Replace CSMA/CA with CSMA/CD because devices can detect collisions after they occur." Incorrect — a transmitting wireless station cannot hear weak signals from hidden terminals, and collisions occur at the receiver, not the sender.

3. TCP Three-Way Handshake — Teaching Diagram + Arithmetic

SIMPLIFIED TEACHING DIAGRAM — BASED ON TEXTBOOK (Tanenbaum Ch. 6) AND LECTURE SLIDES



Both hosts contribute fresh initial sequence numbers. The ACK number is always the next expected byte (ISN + 1 for the SYN, which consumes one sequence number).

Worked arithmetic (exam style)

Given ISN_A = 1543, ISN_B = 2780:
• B's SYN-ACK: Sequence = 2780, Acknowledgement = 1544
• A's final ACK: Sequence = 1544, Acknowledgement = 2781

Given ISN_Student = 4250, ISN_Server = 7600:
• SYN-ACK: seq = 7600, ack = 4251
• Final ACK: seq = 4251, ack = 7601

4. Manual VLSM — Full Worked Example (Past-Paper Style)

Parent block 10.10.0.0/16. Host requirements (descending): LAN1 300, LAN3 100, LAN4 55, LAN2 30, WAN1 2.

Step 1 — Host bits & prefix

300 → 2^9–2 = 510 → /23
100 → 2^7–2 = 126 → /25
55 → 2^6–2 = 62 → /26
30 → 2^5–2 = 30 → /27
2 → 2^2–2 = 2 → /30

Step 2 — Allocation table

Subnet	Network	First usable	Last usable	Broadcast
LAN1 /23	10.10.0.0	10.10.0.1	10.10.1.254	10.10.1.255
LAN3 /25	10.10.2.0	10.10.2.1	10.10.2.126	10.10.2.127
LAN4 /26	10.10.2.128	10.10.2.129	10.10.2.190	10.10.2.191
LAN2 /27	10.10.2.192	10.10.2.193	10.10.2.222	10.10.2.223
WAN1 /30	10.10.2.224	10.10.2.225	10.10.2.226	10.10.2.227

First unallocated address: 10.10.2.228

Technician trap: Assigning 10.10.2.150 to LAN3 is wrong — that address falls inside LAN4 (10.10.2.128/26).

5. Mobile IP — Direct vs Indirect Routing

SIMPLIFIED TEACHING SUMMARY — BASED ON TANENBAUM §5.6.9 AND LECTURE SLIDES (MOBILE NETWORKS)

Components: Home Address (permanent), Care-of Address (temporary on visited network), Home Agent (on home network), Foreign Agent (on visited network, optional).

Indirect (triangle) routing: Correspondent → Home Agent → tunnel to Care-of Address → Mobile. Mobile replies directly to Correspondent. Transparent to the correspondent but adds latency (triangle path).

Direct routing: Correspondent first learns the Care-of Address, then sends packets directly. More efficient; correspondent must be mobility-aware.

Handoff: Process that transfers an active connection from one base station / access point to another while preserving the ongoing session.

6. Final Revision Checklist (tick when confident)

- ■ OSI Layer 1–4 troubleshooting: I can assign the correct layer and one function for each symptom.
- ■ Topologies: I can recommend Star + L2 switches for departmental segmentation and explain why.
- ■ Media: fibre vs microwave over mountains — three justifying characteristics.
- ■ Switching: circuit for real-time voice, packet for bulk transfer; why message switching is wrong for voice.
- ■ Error control: parity detects, Hamming can correct; CRC is strong detection.
- ■ Hidden Terminal + RTS/CTS: I can draw it and explain why CSMA/CD fails on wireless.
- ■ VLSM: I can allocate any past-paper block descending and list Network/First/Last/Broadcast.
- ■ TCP handshake: I can compute every Sequence and ACK number from given ISNs.
- ■ Ports: 80, 443, 22, 25, 53, 21 memorised.
- ■ Mobile IP: Home/Foreign Agent, Care-of Address, triangle vs direct routing.
- ■ CIA triad + one control for each + DoS targets Availability.
- ■ Asymmetric crypto: encrypt with recipient’s public key, decrypt with recipient’s private key.

Version 2 adds the required original textbook figure (OSI) and clearly labelled simplified teaching diagrams for Hidden Terminal, TCP handshake and Mobile IP. Additional textbook page extractions (ALOHA, CSMA figures from Tanenbaum Ch. 4) are available in the figures folder for further insertion if needed. Work every past-paper VLSM and handshake question under timed conditions.